



What Makes Us Unwittingly Share Climate Misinformation Online?

Abstract

Social media is full of false claims about climate change, and even well-meaning users can spread them. A 2025 study co-authored by Harvard Chan C-CHANGE visiting scholar Muhammad Ittefaq tested how 320 people judged false climate posts. It found that posts that looked expert, long, or popular seemed more believable and were more likely to be shared, and that people sometimes shared posts they did not even believe.

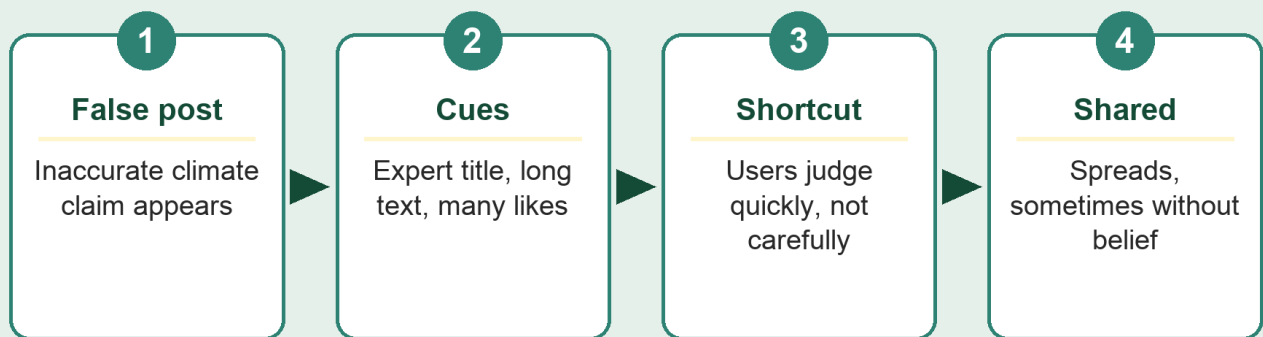
Introduction

Climate change is one of the most important challenges facing the planet, yet social media platforms are filled with climate misinformation. When false or inaccurate messages spread online, they can undermine support for environmental policies, cause political inaction, and increase public doubt about science. Most social media users are not climate experts, so they rely on quick mental shortcuts to decide what to trust. Researchers at Arizona State University, the University of South Carolina, and Harvard Chan C-CHANGE wanted to understand exactly which shortcuts lead people to believe and share false climate information, so that platforms and educators can build better defenses.

How Do Scientists Study This?

The researchers ran an experiment with 320 participants. Each person viewed X-style social media posts containing false climate information. The team varied three cues. Professional expertise cues made a post appear to come from an expert, using titles like Dr. or PhD. Message style cues changed whether a post was long and clearly written or short. Bandwagon cues made a post look popular, with many likes and shares. After viewing each post, participants rated how credible it seemed and how likely they were to share it, on a scale from 1 to 7. The researchers compared ratings across cues, including combinations of two or three cues together.

How misinformation gets shared



Create posts

Researchers made false climate posts that varied expertise, length, and popularity cues.

Rate them

320 participants rated each post's credibility and their intention to share it, from 1 to 7.

Compare cues

Scores were compared to see which cues, alone or combined, made misinformation more persuasive.

What Did They Discover?

All three cues made false posts more persuasive. Posts that seemed to come from an expert rose in credibility from 3.70 to 4.72 out of 7. Longer posts rose from 3.47 to 4.94. Posts that looked popular rose from 3.45 to 4.97, and sharing intention climbed from 3.13 to 4.94. Combining two cues had even bigger effects: a long and popular post raised sharing intention from 3.88 to 5.57. Popularity especially boosted short posts, raising sharing intention from 2.37 to 4.32. Surprisingly, combining all three cues did not increase ratings further. And some participants said they would share a post even if they did not believe it, simply because many others had shared it.

How cues change sharing

320

participants rated false climate posts

3.13→4.94

sharing intention when a post looked popular

3.88→5.57

sharing when a post was long and popular

People do not need to believe a claim to spread it. Popularity alone can push users to pass along misinformation.

Takeaways

Shortcuts mislead

Expert titles, long posts, and high likes make false claims seem more believable.

Popularity spreads

Posts that look widely shared get passed along, even when people doubt them.

Pause before sharing

Checking sources instead of relying on cues helps stop misinformation.

Key Terms

Misinformation

False or inaccurate information, whether or not the person sharing it knows it is wrong.

Heuristic

A mental shortcut people use to make quick judgments, such as trusting a post because it has many likes.

Bandwagon cue

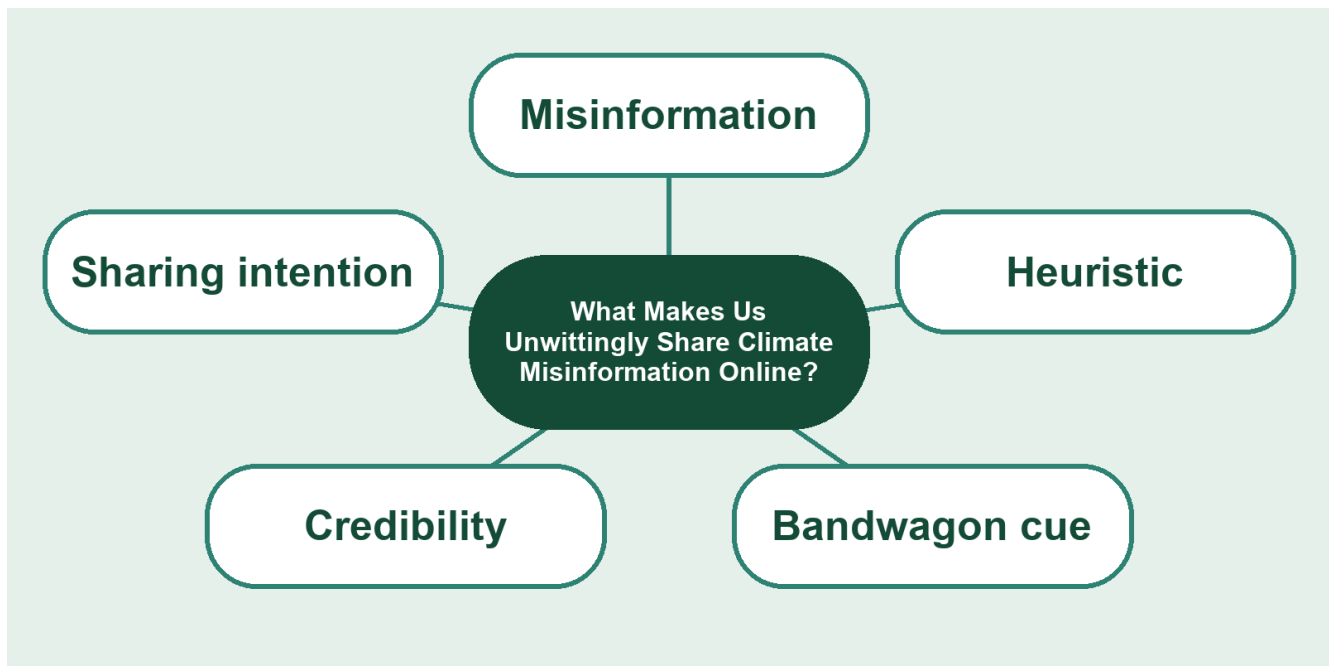
A signal that many other people accept something, like a high number of shares, which makes it seem more trustworthy.

Credibility

How believable or trustworthy a source or message seems.

Sharing intention

How likely a person says they are to pass a message on to others.



Activity

Cue Detective

Students analyze mock social media posts to spot persuasion cues and practice checking claims before sharing.

Materials: Mock social media posts about climate prepared by the teacher, some true and some false; Cue checklist: expertise, length, popularity; Devices for fact-checking (optional)

1. For each post, mark which of the three cues it uses.
2. Rate how credible the post seems from 1 to 7 before checking any facts.
3. Use a reliable source to check the claim, then rate it again.
4. Compare your before and after ratings. Which cue fooled you most?
5. Create a three-step checklist for classmates to use before sharing any climate post.

References

Miller, A. (2025). What makes us (unwittingly) share climate misinformation online? Harvard Chan C-CHANGE.

<https://hsph.harvard.edu/climate-health-c-change/news/what-makes-us-unwittingly-share-climate-misinformation-online/>

Ittefaq, M., et al. (2025). Study in Cogent Social Sciences, as described in the article above.